

Ajitkumar Senthil Kumar

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[Portfolio](#) | [LinkedIn](#) | [GitHub](#)

Summary

Full-stack and AI/ML engineer (2+ years) building **production-grade web applications and APIs** with **FastAPI** and **React/TypeScript**. Experienced in delivering **scalable, cloud-deployed systems** using **Docker, CI/CD (GitHub Actions), SQL**, and modern DevOps practices. Proven at **integrating ML inference and automated data pipelines into live products** to improve system performance, reliability, and user experience.

Education

State University of New York at Buffalo | *M.S., Data Science (GPA: 3.5/4)* **2024 - 2025**

Anna University, Chennai | *B.E., Mechanical Engineering (GPA: 8.1/10)* **2017 - 2021**

Experience

Human CloudSoft | *Software Intern* **May 2023 – Jun 2024**

- Developed a web-based inventory management platform that automated stock tracking, transitioning all inventory records from manual to digital workflows.
- Improved inventory accuracy by 35% and reduced audit time by 40% by implementing automated alerts, validation checks, and streamlined operational processes.
- Designed **REST APIs** and backend endpoints to support real-time updates, role-based access, and reliable transaction handling.

Klicknet Info Services Pvt Ltd | *Frontend Developer* **Apr 2022 – Apr 2023**

- Engineered responsive, accessible React and TypeScript interfaces while collaborating with backend teams to optimize workflows, increasing user retention by 25%, improving cross-team efficiency by 30%, and delivering scalable, reusable components that accelerated feature development and improved maintainability.
- Translated UX designs into pixel-accurate, cross-device interfaces while maintaining accessibility, performance, and consistent frontend architecture standards.

L&T Rubber Processing Machinery | *Design Engineer* **Nov 2021–Feb 2022**

- Converted complex 3D models into precise 2D AutoCAD drawings to ensure production readiness and defect-free outputs.
- Reduced manufacturing defects by 15% and rework by 20% through precision CAD modeling in CREO and application of GD&T principles.

Projects

SmartLoanPred | *Full-stack ML & AI Application* **Sep 2025–Present**

- Built and deployed a **full-stack, real-time loan approval system** (classification and regression) using React, Tailwind, and FastAPI with JWT/OAuth2 role-based access, Dockerized CI/CD (GitHub Actions), NGINX reverse proxy, and integrated LLM reports with a **RAG-based loan assistant** for contextual loan Q&A.

MIDAS | *AI-Powered Medical Imaging & Clinical Risk Assessment Platform* **Jul 2025–Present**

- Built end-to-end computer vision pipelines in PyTorch for dermatology risk screening and respiratory disease detection from medical images, leveraging extensive preprocessing, augmentation, and class-weighted loss to improve performance across diverse datasets.
- Productionized Machine Learning Inference Services by deploying TorchScript/ONNX models via FastAPI and integrating **Grad-CAM** for explainability, complemented by an **Agentic RAG** system to generate scoped, clinician-facing medical insights.

New York Taxi Rides | *MLOps* **Feb 2024–Apr 2024**

- Modeled NYC taxi ride demand using historical trip data with extensive feature engineering (lag variables, rolling windows, external joins with holidays and weather), validated using time-based cross-validation, and trained models on AWS SageMaker. Integrated features into a Hopworks feature store, **stored artifacts in S3**, triggered **automated retraining via AWS Lambda function**, and built a Streamlit dashboard for real-time monitoring with CI/CD workflows orchestrated using GitHub Actions.

Technical Skills

- Programming:** Python, R, SQL, JavaScript, TypeScript
- Frontend:** React, Next.js, Tailwind CSS, shadcn/ui
- Backend:** FastAPI, SQLAlchemy, Firebase(OAuth)
- ML/AI:** Scikit-learn, PyTorch, Pandas, NumPy, imbalanced-learn, MLflow, Jupyter, LangChain, FAISS/Chroma, Ollama
- Databases/Cloud:** PostgreSQL, SQL Server, Supabase, SQLite; Docker, GitHub Actions, AWS (SageMaker,S3,Lambda).